Modern AI Engineering

Phase 6 — Full Course Guide

NLP · Computer Vision · LLMs · Generative AI · RAG · Vector Databases · Agentic AI · Prompt Engineering · FastAPI

Each topic has 3 courses ranked in the order you should take them. Course #1 is the best starting point and will take you from beginner to advanced on its own. Courses #2 and #3 go deeper. Work through the topics in the numbered sequence — the order is intentional and each topic builds on the one before it.

Why this order?

- NLP first — transformers, BERT, and sequence modelling are the foundation that LLMs are built on. Learning NLP before LLMs means the architecture makes sense rather than feeling like magic.
- Computer Vision second — uses the same deep learning foundations (CNNs, attention, transformers) applied to images. Having both NLP and CV done before GenAI means diffusion models and multimodal AI will click properly.
- LLMs third — with NLP and CV as foundations, LLMs are a natural progression. No black box.
- Generative AI fourth — you now have the architecture knowledge to understand training, alignment, and diffusion properly.
- RAG and Vector Databases fifth and sixth — these only make sense once you know what an LLM is and why it needs external memory.
- Agents seventh — the most complex topic. Agents combine LLMs + tools + RAG + memory. You now have every prerequisite.
- Prompt Engineering and Fine-Tuning eighth — you know how LLMs work, now you learn to control and customise them precisely.
- FastAPI last — the deployment layer. You now have a complete skill set to expose as production APIs.
- System Design — only makes sense after you have built real systems — APIs, LLM pipelines, RAG systems, agents. After Phase 6 you have built all of those. Now you learn to design them at scale. Before Phase 6 it is too abstract to stick.

Quick Reference — All Courses at a Glance

📝 6.1 — Natural Language Processing (NLP)			
#1	**CS224n: NLP with Deep Learning** *Stanford University*	Intermediate → Advanced	web.stanford.edu/class/cs224n/
#2	**HuggingFace NLP Course** *Hugging Face*	Beginner → Advanced	huggingface.co/learn/nlp-course
#3	Stanford CS336: Language Modeling from Scratch	Beginner → Advanced	https://www.youtube.com/playlist?list=PLoROMvodv4rOY23Y0BoGoBGgQ1zmU_MT
#4	**Practical Natural Language Processing (Book)** *O'Reilly — free PDF*	Beginner → Advanced	www.practicalnlp.ai/

👁 6.2 — Computer Vision			
#1	**CS231n: CNNs for Visual Recognition** *Stanford University*	Intermediate → Advanced	cs231n.github.io/
#2	**Practical Deep Learning for Coders** *fast.ai*	Beginner → Advanced	course.fast.ai/
#3	**OpenCV Python Full Course** *Murtaza's Workshop · YouTube*	Beginner → Intermediate	www.youtube.com/watch?v=WQeoO7MI0Bs
#4	**WQU Computer Vision Projects** *World Quant University*	Capstone Projects	https://www.wqu.edu/computer-vision-lab

🧠 6.3 — Large Language Models (LLMs)			
#1	**Karpathy: Deep Dive into LLMs like ChatGPT** YouTube	Beginner → Advanced	https://www.youtube.com/watch?v=7xTGNNLPyMI
#2	**LLM Zoomcamp** *DataTalksClub*	Beginner → Advanced	github.com/DataTalksClub/llm-zoomcamp
#3	Maxime Labonne: LLM Course **(GitHub)**	Beginner → Advanced	https://github.com/mlabonne/llm-course
#4	**Hugging Face LLM Course (Updated)**	Beginner → Advanced	https://huggingface.co/learn/llm-course
#5	**CS324 — Large Language Models** *Stanford University*	Intermediate → Advanced	stanford-cs324.github.io/winter2022/
#6	**Hands-On Large Language Models (Book + Notebooks)** *Paul Iusztin et al. · GitHub*	Intermediate → Advanced	github.com/HandsOnLLM/Hands-On-Large-Langu

✨ 6.4 — Generative AI			
#1	MIT 6.S191: Deep Generative Modelling Lecture	Beginner → Advanced	https://www.youtube.com/playlist?list=PLtBw6njQRU-rwp5__7C0oIVt26ZgjG9NI
#2	**Generative AI with LLMs** *DeepLearning.AI + AWS · Coursera (audit free)*	Beginner → Advanced	www.coursera.org/learn/generative-ai-with-
#3	**Full Stack LLM Bootcamp** *The Full Stack*	Intermediate → Advanced	fullstackdeeplearning.com/llm-bootcamp/
#4	**How Diffusion Models Work** *DeepLearning.AI + Hugging Face (free)*	Intermediate → Advanced	www.deeplearning.ai/short-courses/how-diff
#5	Diffusion Models Course *Hugging Face*	Intermediate → Advanced	https://huggingface.co/learn/diffusion-course/en/unit0/1

🔍 6.5 — RAG — Retrieval-Augmented Generation			
#1	**RAG from Scratch** *LangChain + freeCodeCamp · YouTube*	Beginner → Advanced	www.youtube.com/watch?v=sVcwVQRHIc8
#2	Cohere: RAG Production Guide *University*	Intermediate → Advanced	https://docs.cohere.com/docs/retrieval-augmented-generation-rag
#3	**Building and Evaluating Advanced RAG** *DeepLearning.AI + LlamaIndex (free)*	Intermediate → Advanced	www.deeplearning.ai/short-courses/building
#4	**Production RAG Tutorials** Deepset *Haystack creators*	Intermediate → Advanced	https://haystack.deepset.ai/tutorials
#3	**Building Agentic RAG with LlamaIndex** *DeepLearning.AI + LlamaIndex (free)*	Advanced	www.deeplearning.ai/short-courses/building

🗄 6.6 — Vector Databases			
#1	**Vector Databases: from Embeddings to Applications** *DeepLearning.AI + Weaviate (free)*	Beginner → Intermediate	www.deeplearning.ai/short-courses/vector-d
#2	**ChromaDB Full Tutorial** *freeCodeCamp · YouTube*	Beginner → Intermediate	www.youtube.com/watch?v=3yPBVii7Ct0
#3	**Pinecone Learning Centre** *Pinecone (official · free)*	Beginner → Advanced	www.pinecone.io/learn/

🤖 6.7 — Agentic AI			
#1	**AI Agents in LangGraph**	Intermediate →	www.deeplearning.ai/short-

	DeepLearning.AI + LangChain (free)	Advanced	courses/ai-agent
#2	**AI Agentic Design Patterns with AutoGen** *DeepLearning.AI + Microsoft (free)*	Intermediate → Advanced	www.deeplearning.ai/short-courses/ai-agent
#3	**LangChain for LLM Application Development** *DeepLearning.AI + LangChain (free)*	Beginner → Intermediate	www.deeplearning.ai/short-courses/langchai
#4	**CrewAI Full Course** Brandon Hancock YouTube	Beginner → Intermediate	https://www.youtube.com/watch?v=sPzc6hMg7So
#5	**Agentic AI Developer Certification** *Ready Tensor*	Intermediate → Advanced	https://app.readytensor.ai/hubs/ready_tensor_certifications
#6	**Cookbook: Agents (Reference)** *OpenAI · GitHub*	Beginner → Intermediate	https://github.com/openai/openai-cookbook/tree/main/examples/agents_sdk

⚙️ 6.8 — Prompt Engineering + Fine-Tuning			
#1	**ChatGPT Prompt Engineering for Developers** *DeepLearning.AI + OpenAI (free)*	Beginner → Intermediate	www.deeplearning.ai/short-courses/chatgpt-
#2	**Fine-Tuning Large Language Models** *DeepLearning.AI (free)*	Intermediate → Advanced	www.deeplearning.ai/short-courses/finetuni
#3	**PEFT Documentation** *Hugging Face (free)*	Advanced	huggingface.co/docs/peft/index

🚀 6.9 — FastAPI for AI Services			
#1	**FastAPI Full Course** *Bitfumes · YouTube*	Beginner → Intermediate	www.youtube.com/watch?v=tLKKmouUams
#2	**FastAPI Official Documentation** *fastapi.tiangolo.com*	Beginner → Advanced	fastapi.tiangolo.com/
#3	**Serving ML Models with FastAPI** *TestDriven.io (free articles)*	Intermediate	testdriven.io/blog/fastapi-machine-learnin

Full Course Descriptions

Top 3 courses per topic — in the order you should study them

6.1 📝 Natural Language Processing (NLP)

Transformers · BERT · GPT · Text Classification · NER · Summarisation

📍 *Start here. NLP builds your understanding of transformers, attention, and language models. Everything in LLMs (6.3) and GenAI (6.4) will make far more sense because you will already understand the architecture they are built on.*

#1 — Best Overall — CS224n: NLP with Deep Learning

Stanford University · Intermediate → Advanced · Full semester · self-paced

🔗 https://web.stanford.edu/class/cs224n/

What you will learn

- Word vectors: Word2Vec, GloVe, FastText — from scratch
- Neural network classifiers for text
- RNNs, LSTMs, GRUs for sequence modelling
- Attention mechanisms — building up to transformers
- Transformer architecture in complete detail
- BERT, GPT, T5 — pre-training objectives and architectures
- Machine translation, summarisation, question answering
- Coreference resolution and dependency parsing

✅ **Strengths:** The gold standard NLP course globally. Christopher Manning co-invented many of the techniques taught. Full lecture videos free on YouTube.

🎯 **Best for:** Anyone who wants the deepest possible theoretical NLP foundation before moving to LLMs.

#2 — HuggingFace NLP Course

Hugging Face · Beginner → Advanced · 8 chapters · self-paced

🔗 https://huggingface.co/learn/nlp-course

What you will learn

- Transformers library: pipelines, tokenisers, models
- Fine-tuning BERT on custom datasets
- Text classification, NER, question answering, summarisation
- Sharing models and datasets on the HuggingFace Hub
- Efficient training: mixed precision, gradient accumulation
- Token classification and sequence-to-sequence tasks

✅ **Strengths:** Built by the people who created the tools. Every concept has a runnable notebook. The bridge between NLP theory and the actual industry standard library.

🎯 **Best for:** Turning NLP knowledge into production-ready code for every major NLP task.

#3 — Stanford CS336: Language Modeling from Scratch (2025)

Stanford University · Full 2025 Course · Free on YouTube

🔗 https://www.youtube.com/playlist?list=PLoROMvodv4rOY23Y0BoGoBGgQ1zmU_MT_

What you will learn

- Building a full LLM pipeline from raw data to trained model
- Data collection, cleaning, and deduplication at scale
- Tokenisation and vocabulary building
- Transformer architecture implementation from scratch
- Pretraining at scale — compute budgets and scaling laws
- Fine-tuning and instruction following
- Evaluation benchmarks and RLHF

✅ **Strengths:** Free full book PDF. Covers the widest range of real-world NLP tasks. Great for applied NLP across industry domains.

🎯 **Best for:** Applied NLP practitioners who need broad coverage of real-world use cases.

#4 — Practical Natural Language Processing (Book)

O'Reilly — free PDF · Beginner → Advanced · 12 chapters · self-paced

🔗 https://www.practicalnlp.ai/

What you will learn

- Text preprocessing pipeline from raw data to features
- Classical ML for NLP: Naive Bayes, SVMs, logistic regression
- Deep learning for NLP: CNNs, RNNs, attention
- Information extraction: NER, relation extraction
- Text classification and sentiment analysis end-to-end
- Building chatbots and dialogue systems
- NLP for specific domains: biomedical, social media, multilingual
- Deployment of NLP pipelines

✅ **Strengths:** Free full book PDF. Covers the widest range of real-world NLP tasks. Great for applied NLP across industry domains.

Use this as a reference book alongside Steps 1–3. It is the widest-coverage NLP resource — biomedical NLP, social media, multilingual, chatbots. Read chapters relevant to whatever you are building. Not a primary course — a companion.

🎯 **Best for:** Applied NLP practitioners who need broad coverage of real-world use cases.

6.2 👁 Computer Vision

CNNs · Object Detection · Segmentation · OpenCV · Real-Time Applications

📍 *Study CV after NLP. Both use the same deep learning foundations — attention, transformers, CNNs — but applied to images. Having NLP first means you already understand the building blocks. Doing both before LLMs and GenAI means you will understand multimodal AI and diffusion models properly.*

#1 — Best Overall — CS231n: CNNs for Visual Recognition

Stanford University · Intermediate → Advanced · Full 10-week semester · self-paced

🔗 https://cs231n.github.io/

What you will learn

- Image classification: kNN, linear classifiers, SVMs
- Neural network foundations and backpropagation
- Convolutional networks: architecture, visualisation, transfer learning
- Training in practice: data augmentation, batch norm, dropout
- Object detection: R-CNN, YOLO, SSD
- Image segmentation: semantic, instance, panoptic
- Generative models for vision: VAEs, GANs, Diffusion
- Video understanding and 3D vision

✅ **Strengths:** The definitive CV course. Taught at Stanford since 2015. Lecture notes are possibly the best-written CV material in the world. Completely free.

🎯 **Best for:** Anyone who wants a world-class rigorous foundation in computer vision.

#2 — Practical Deep Learning for Coders

fast.ai · Beginner → Advanced · ~7 weeks · self-paced

🔗 https://course.fast.ai/

What you will learn

- Image classification with state-of-the-art accuracy from Day 1
- Data augmentation, transfer learning, fine-tuning
- Multi-label classification and image segmentation
- Object detection with YOLO and RetinaNet
- Training on small datasets — the fast.ai specialty
- Deploying CV models to web apps and mobile
- Diffusion models for image generation

✅ **Strengths:** Jeremy Howard's top-down teaching style — build first, understand later. You will have working models on Day 1. Unmatched for getting practical results quickly.

🎯 **Best for:** Getting results fast. Best if theory-first approaches feel slow or demotivating.

#3 — OpenCV Python Full Course

Murtaza's Workshop · YouTube · Beginner → Intermediate · ~4 hours

🔗 https://www.youtube.com/watch?v=WQeoO7MI0Bs

What you will learn

- Core OpenCV operations: reading, resizing, transforming images
- Colour spaces and colour detection
- Contour detection, shape recognition, drawing
- Face detection with Haar cascades
- Real-time video processing
- Tracking objects in video
- Document scanning pipeline

✅ **Strengths:** Pure hands-on. Every concept shown in working code immediately. The best course for learning the OpenCV library itself.

🎯 **Best for:** Building practical CV applications and real-time tools.

#4 — WQU Computer Vision Projects

World Quant University · Beginner → Advanced · self-paced

🔗 https://www.wqu.edu/computer-vision-lab

What you will learn

- Build computer vision projects

✅ **Strengths:** This gives you the ability to build massive computer vision projects for your portfolio

🎯 **Best for:** Anyone who wants to apply a world-class rigorous foundation in computer vision.

6.3 🧠 Large Language Models (LLMs)

Architecture · Training · Fine-Tuning · Deployment

📍 *Now that you understand transformers from CS224n and deep networks from CS231n, LLMs will click immediately rather than feeling like a black box. You are ready for the full depth of this topic.*

#1 — Karpathy: Deep Dive into LLMs like ChatGPT

Free YouTube · 3+ hours · · Beginner → Advanced · 12 chapters · self-paced

🔗 https://github.com/HandsOnLLM/Hands-On-Large-Language-Models

Watch this first.

- It is a 3-hour masterclass where Karpathy walks through the entire ChatGPT pipeline. Pretraining, supervised fine-tuning, RLHF, inference — with complete clarity.

  It is not a course; it is a mental map of everything you are about to learn. After this video, every other LLM resource will feel organised and logical instead of scattered.

✅ **Strengths:** **G**ives solid foundation in LLM

#2 — Best Overall — LLM Zoomcamp

DataTalksClub · Beginner → Advanced · 10 weeks · self-paced

🔗 https://github.com/DataTalksClub/llm-zoomcamp

What you will learn

- Transformer architecture and how LLMs work internally
- Prompt engineering and prompt chaining
- Building RAG systems from scratch
- Fine-tuning open-source LLMs — Mistral, LLaMA
- Evaluation frameworks — RAGAS, LLM-as-judge
- Deploying LLM applications with Docker and cloud
- Monitoring and observability for LLM systems

✅ **Strengths:** The most complete, production-grade, fully free LLM course available. Real weekly projects — not just videos.

🎯 **Best for:** Anyone who wants to build with LLMs in production, not just understand them conceptually.

#3 Maxime Labonne: LLM Course (GitHub)

Hugging Face researcher · Free · Full roadmap with Colab notebooks

🔗 https://github.com/mlabonne/llm-course

What you will learn

- Do alongside or immediately after LLM Zoomcamp.

This GitHub repository is the most comprehensive free LLM curriculum available. It covers the LLM Scientist track (how to build and train LLMs) and the LLM Engineer track (how to deploy and use them).
Every topic has a runnable Colab notebook. Topics include quantisation (GGUF, GPTQ, AWQ), model merging, preference alignment (DPO, ORPO), and production deployment patterns — things

most courses skip.

#4 Hugging Face LLM Course (Updated)

Hugging Face · Free · No ads

🔗 https://huggingface.co/learn/llm-course

What you will learn

- Do after LLM Zoomcamp

The HuggingFace NLP course has been fully rebuilt as an LLM course. Chapters 1–9 cover transformers and fine-tuning. Chapters 10–12 go into advanced topics: fine-tuning for reasoning, curating high-quality datasets, building reasoning models. Everything has runnable notebooks.

#5 — CS324 — Large Language Models

Stanford University · Intermediate → Advanced · Full semester · self-paced

🔗 https://stanford-cs324.github.io/winter2022/

What you will learn

- Deep theoretical understanding of LLM architecture
- Training dynamics, scaling laws, and emergent abilities
- Alignment, safety, and RLHF
- In-context learning and chain-of-thought reasoning
- Evaluation methodologies and benchmarks
- Societal impacts and ethics of LLMs

✅ **Strengths:** Stanford-level rigour. Lecture notes are among the best LLM reading material anywhere. Excellent for understanding why things work.

🎯 **Best for:** Building deep theoretical foundations alongside practical work.

#6 — Hands-On Large Language Models (Book + Notebooks)

Paul Iusztin et al. · GitHub · Intermediate → Advanced · 12 chapters · self-paced

🔗 https://github.com/HandsOnLLM/Hands-On-Large-Language-Models

What you will learn

- Text representation and embedding models

- Semantic search and dense retrieval
- LLM architecture from scratch
- Fine-tuning with QLoRA and LoRA
- RAG pipelines with vector databases
- Advanced generation techniques: beam search, sampling
- Production deployment patterns

✅ **Strengths:** Free full book with runnable notebooks. O'Reilly quality available for free. Visual and code-forward throughout.

🎯 **Best for:** People who learn best by reading and running code simultaneously.

6.4 ✨ Generative AI

RLHF · Fine-Tuning · Diffusion Models · Full-Stack GenAI Engineering

📍 *After LLMs you have the architecture knowledge. Now you learn how to train, align, and deploy generative systems — including image generation with diffusion models, which draws on your CV foundation from 6.2.*

#1 MIT 6.S191: Deep Generative Modelling Lecture MIT

Free on YouTube · ~1.5 hours · Watch first

🔗 https://www.youtube.com/playlist?list=PLtBw6njQRU-rwp5__7C0oIVt26ZgjG9NI

What you will learn

- Watch this lecture before anything else in 6.4.

It gives you a clear, clean conceptual map of VAEs, GANs, and diffusion models in under 2 hours. Once you have this mental picture, the other courses in 6.4 will feel structured rather than overwhelming.

#2 — Best Overall — Generative AI with LLMs

DeepLearning.AI + AWS · Coursera (audit free) · Beginner → Advanced · ~3 weeks

🔗 https://www.coursera.org/learn/generative-ai-with-llms

What you will learn

- How generative AI works: transformers, pre-training, inference
- Prompt engineering and in-context learning
- Fine-tuning: full fine-tuning, PEFT, LoRA, QLoRA
- Reinforcement Learning from Human Feedback (RLHF)
- Constitutional AI and alignment techniques
- Model optimisation: distillation, quantisation, pruning
- Building GenAI-powered applications end-to-end

✅ **Strengths:** The most complete structured GenAI course available for free. Covers RLHF and alignment in a way almost no other free course does. Co-taught by AWS engineers.

🎯 **Best for:** A complete, structured GenAI education from architecture to production deployment.

#3 — Full Stack LLM Bootcamp

The Full Stack · Intermediate → Advanced · 8 lectures · self-paced

🔗 https://fullstackdeeplearning.com/llm-bootcamp/

What you will learn

- The full GenAI engineering stack: model to product
- Prompt engineering: few-shot, chain-of-thought, system prompts
- LLM operations: evaluation, monitoring, cost management
- Augmented language models — tools, retrieval, memory
- UX patterns for LLM-powered products
- The complete deployment pipeline for a GenAI application
- What makes GenAI projects succeed vs fail in production

✅ **Strengths:** Taught by engineers who built real AI products. The best resource for bridging ML research and product engineering.

🎯 **Best for:** Engineers who want to ship GenAI products, not just understand the theory.

#4 — How Diffusion Models Work

DeepLearning.AI + Hugging Face (free) · Intermediate → Advanced · ~2 hours

🔗 https://www.deeplearning.ai/short-courses/how-diffusion-models-work/

What you will learn

- How diffusion models add and progressively remove noise
- U-Net architecture for image generation
- Training a diffusion model from scratch
- DDPM — Denoising Diffusion Probabilistic Models
- Classifier-free guidance for controlling generation output
- Context conditioning — generating from text prompts
- Stable Diffusion internals: CLIP, VAE, U-Net

✅ **Strengths:** The best free course for understanding AI image generation technically. You build a diffusion model from scratch. Your CV background from 6.2 makes this click immediately.

🎯 **Best for:** Understanding how Stable Diffusion, DALL-E, and Midjourney actually work.

#5 Hugging Face: Diffusion Models Course Hugging Face

Free · No paywall

🔗 https://huggingface.co/learn/diffusion-course/en/unit0/1

What you will learn

- Do after Step 4

This goes deeper into diffusion on the HuggingFace tooling side — fine-tuning diffusion models, guidance techniques, building your own pipeline. Step 4 gives you the concepts; this gives you the implementation tools.

6.5 🔍 RAG — Retrieval-Augmented Generation

Building · Evaluating · Optimising RAG Pipelines

📍 *RAG only makes sense once you understand LLMs — it is the method for giving LLMs access to your own knowledge. You now have the right foundation to go deep on this.*

#1 — Best Overall — RAG from Scratch

LangChain + freeCodeCamp · YouTube · Beginner → Advanced · ~4 hours + notebooks

🔗 https://www.youtube.com/watch?v=sVcwVQRHIc8

What you will learn

- What RAG is and the core problem it solves
- Document loading, splitting, and chunking strategies
- Embedding models and similarity search
- Vector stores: ChromaDB, FAISS, Pinecone
- Basic → Advanced → Modular RAG
- Query translation, routing, and re-ranking
- Multi-vector retrieval and parent document retrieval
- RAG evaluation with RAGAS framework

✅ **Strengths:** The most comprehensive free RAG resource. Built by the LangChain team. All notebooks included.

🎯 **Best for:** The best first RAG resource. Start here.

#2 Cohere: RAG Production Guide Cohere LLM University

Free .

🔗 https://docs.cohere.com/docs/retrieval-augmented-generation-rag

What you will learn

- Do after Step 1

Where Step 1 teaches you to build RAG, this teaches you the production concerns — latency, cost, retrieval quality, re-ranking strategies. Cohere's documentation goes deeper into dense retrieval and real-world performance than most tutorials.

#3 — Building and Evaluating Advanced RAG

DeepLearning.AI + LlamaIndex (free) · Intermediate → Advanced · ~2 hours

🔗 https://www.deeplearning.ai/short-courses/building-evaluating-advanced-rag/

What you will learn

- Sentence-window retrieval for better context capture
- Auto-merging retrieval for hierarchical documents
- Advanced indexing strategies with LlamaIndex
- RAG evaluation triad: context relevance, groundedness, answer relevance
- Using TruLens for RAG monitoring in production

✅ **Strengths:** Taught by LlamaIndex creators. Focuses on quality — how to measure and systematically improve RAG performance.

🎯 **Best for:** After Course #1 — making your RAG pipeline production-quality.

#4 Haystack: Production RAG Tutorials Deepset (Haystack creators)

Free .

🔗 https://haystack.deepset.ai/tutorials

What you will learn

- Do after Step 3

Haystack is the production-grade RAG framework used by enterprises. These tutorials cover what LangChain tutorials leave out — multi-hop retrieval, conversational RAG, hybrid pipelines, and enterprise evaluation. This is the gap between "it works on my laptop" and "it works in production."

#5 — Building Agentic RAG with LlamaIndex

DeepLearning.AI + LlamaIndex (free) · Advanced · ~2 hours

🔗 https://www.deeplearning.ai/short-courses/building-agentic-rag-with-llamaindex/

What you will learn

- Router agents that choose between data sources intelligently
- Tool-calling agents that query documents like databases
- Multi-document agents with cross-document reasoning
- Combining RAG and agents into unified pipelines
- Agentic research assistants that plan their own retrieval
- Debugging and evaluating agentic RAG systems

✅ **Strengths:** The bridge between RAG and agents — covered nowhere else for free.

🎯 **Best for:** After Courses #1 and #2 — combining retrieval with autonomous reasoning.

6.6 🗄 Vector Databases

Embeddings · ANN Search · ChromaDB · Weaviate · Pinecone · Hybrid Search

📍 *Vector databases are the engine that makes RAG work. Now that you have built RAG pipelines, you need to understand what is happening inside the database — HNSW, ANN search, hybrid retrieval — so you can make smart architectural decisions.*

#1 — Best Overall — Vector Databases: from Embeddings to Applications

DeepLearning.AI + Weaviate (free) · Beginner → Intermediate · ~2 hours

🔗 https://www.deeplearning.ai/short-courses/vector-databases-embeddings-applications/

What you will learn

- What embeddings are and why they capture semantic meaning
- ANN algorithms: HNSW, IVF, PQ — how they work

- How vector databases differ from traditional SQL databases
- Weaviate in depth: schema design, import, querying
- Sparse vs dense retrieval: keyword vs semantic search
- Hybrid search: combining BM25 and vector search
- Multi-modal vector search — text and image in one index

✅ **Strengths:** The most comprehensive free standalone vector database course. Taught by Weaviate engineers.

🎯 **Best for:** Understanding vector databases from first principles — not just using them as a black box.

#2 — ChromaDB Full Tutorial

freeCodeCamp · YouTube · Beginner → Intermediate · ~2.5 hours

🔗 https://www.youtube.com/watch?v=3yPBVii7Ct0

What you will learn

- ChromaDB from zero: collections, documents, embeddings
- Embedding functions: OpenAI, HuggingFace, local models
- Querying by similarity, metadata filtering, hybrid search
- Persisting and loading ChromaDB collections
- Integrating ChromaDB with LangChain for RAG
- Multi-modal ChromaDB with image embeddings

✅ **Strengths:** The most practical ChromaDB tutorial. ChromaDB is the primary local vector database for personal and small-scale projects.

🎯 **Best for:** Getting ChromaDB working in your own projects immediately.

#3 — Pinecone Learning Centre

Pinecone (official · free) · Beginner → Advanced · 20+ articles · self-paced

🔗 https://www.pinecone.io/learn/

What you will learn

- Vector similarity search from first principles
- Dense vs sparse vectors and when to use each
- Pinecone architecture: pods, indexes, namespaces
- Semantic search at production scale
- Hybrid sparse-dense retrieval
- Recommendation systems with vector search
- Comparison: ChromaDB vs Weaviate vs Qdrant vs Pinecone

✅ **Strengths:** The best free resource for production-scale vector search. Widely considered the clearest explanation of vector databases available.

🎯 **Best for:** Production deployments and choosing the right vector database for the job.

6.7 🤖 Agentic AI

Agents · Tool Use · Multi-Agent Systems · LangGraph · AutoGen

📍 *Agents are the most complex topic in this guide. They combine LLMs, tools, RAG, and memory into systems that act autonomously. Having completed 6.3 through 6.6 means you now have every prerequisite.*

#1 — Best Overall — AI Agents in LangGraph

DeepLearning.AI + LangChain (free) · Intermediate → Advanced · ~3 hours

🔗 https://www.deeplearning.ai/short-courses/ai-agents-in-langgraph/

What you will learn

- The agent loop: plan, act, observe
- Building stateful multi-step agents with LangGraph
- Tool use: connecting LLMs to APIs, databases, code runners
- ReAct (Reasoning + Acting) agent pattern
- Agent memory: short-term state vs long-term persistence
- Human-in-the-loop agents for critical action approval
- Multi-agent systems where agents coordinate with each other

✅ **Strengths:** Built by the LangGraph team. Most up-to-date agentic course available. LangGraph is the industry standard for production agents.

🎯 **Best for:** Building reliable, production-grade agents.

#2 — AI Agentic Design Patterns with AutoGen

DeepLearning.AI + Microsoft (free) · Intermediate → Advanced · ~3 hours

🔗 https://www.deeplearning.ai/short-courses/ai-agentic-design-patterns-with-autogen/

What you will learn

- 4 core agent design patterns: Reflection, Tool Use, Planning, Multi-Agent
- Agents that review and improve their own output iteratively
- AutoGen: Microsoft's multi-agent platform
- Conversational agents that collaborate on complex problems
- Code-writing and code-executing agents
- Custom agent roles and orchestration patterns

✅ **Strengths:** Teaches agent design patterns — transferable concepts that work beyond any single framework. Andrew Ng teaches the pattern theory personally.

🎯 **Best for:** Understanding agent architecture at a conceptual level, not just implementation.

#3 — LangChain for LLM Application Development

DeepLearning.AI + LangChain (free) · Beginner → Intermediate · ~2 hours

🔗 https://www.deeplearning.ai/short-courses/langchain-for-llm-application-development/

What you will learn

- LangChain fundamentals: chains, prompts, parsers
- Memory types: buffer, summary, entity, vector-based
- Question-answering systems over documents
- Evaluation of LLM applications
- Agents and tools in LangChain
- Building end-to-end LLM applications

✅ **Strengths:** The foundation course before LangGraph. LangChain is the most widely used LLM application framework.

🎯 **Best for:** Understanding the LangChain ecosystem as a foundation before LangGraph agents.

#4 CrewAI Full Course Brandon Hancock

YouTube · Free

🔗 https://www.youtube.com/watch?v=sPzc6hMg7So

What you will learn

- Do after AutoGen

CrewAI is the third major agentic framework. It specialises in building multi-agent teams where each agent has a role, goal, and toolset. Knowing LangGraph, AutoGen, AND CrewAI means you can pick the right tool for any job — and you understand agent architecture from three different angles.

#5 — Ready Tensor: Agentic AI Developer Certification

Ready Tensor · 12 weeks · Project-based · Free to enroll

🔗 https://app.readytensor.ai/hubs/ready_tensor_certifications

What you will learn

- Do this after Steps 1–4.

This is the most comprehensive agentic AI program available. 12 weeks across three modules

- Module 1: agent foundations and RAG;
- Module 2: multi-agent workflows with LangGraph and MCP;
- Module 3: production deployment, guardrails, testing, and monitoring.

Each module ends with a reviewed portfolio project. All lesson content is also on their GitHub. The certificate and badge require the paid tier (affordable, with scholarships on Discord) — but the learning content is free.

#6 OpenAI Cookbook: Agents (Reference)

OpenAI · GitHub · Free · Use as reference throughout

🔗 https://github.com/openai/openai-cookbook/tree/main/examples/agents_sdk

What you will learn

- Not a course — a reference.
-

Production-quality code patterns for tool use, planning, and memory from OpenAI engineers. Use this throughout Steps 2–5 whenever you want to see how something is done at production level.

6.8 ⚙️ Prompt Engineering + Fine-Tuning

Prompt Design · LoRA · QLoRA · PEFT · Instruction Tuning

📍 *You now know how LLMs and agents work. This section teaches you to control and customise them precisely — through prompting for immediate changes and fine-tuning for deeper specialisation.*

#1 — Best Overall — ChatGPT Prompt Engineering for Developers

DeepLearning.AI + OpenAI (free) · Beginner → Intermediate · ~2 hours

🔗 https://www.deeplearning.ai/short-courses/chatgpt-prompt-engineering-for-developers/

What you will learn

- Core principles: clarity, specificity, structured output
- Zero-shot, one-shot, and few-shot prompting
- Chain-of-thought prompting for complex reasoning
- System prompts and role assignment
- Iterative prompt refinement methodology
- Summarisation, transformation, expansion, extraction
- Building a complete customer service chatbot

✅ **Strengths:** Taught by Andrew Ng and Isa Fulford from OpenAI. The definitive free prompt engineering course.

🎯 **Best for:** Learning systematic, principled prompt engineering from the ground up.

#2 — Fine-Tuning Large Language Models

DeepLearning.AI (free) · Intermediate → Advanced · ~2 hours

🔗 https://www.deeplearning.ai/short-courses/finetuning-large-language-models/

What you will learn

- When to fine-tune vs use prompting
- Instruction fine-tuning on custom datasets
- Training loop for fine-tuning LLMs
- Parameter-efficient fine-tuning: LoRA and QLoRA
- Evaluating fine-tuned models
- Preparing datasets for fine-tuning
- Deploying fine-tuned models

✅ **Strengths:** The clearest free introduction to fine-tuning. Addresses the practical decision of when fine-tuning is actually worth the cost.

🎯 **Best for:** Understanding when and how to fine-tune LLMs for specific tasks.

#3 — PEFT Documentation

Hugging Face (free) · Advanced · Self-paced reference

🔗 https://huggingface.co/docs/peft/index

What you will learn

- LoRA: low-rank adaptation in full technical detail
- QLoRA: quantised fine-tuning on consumer hardware
- Adapter layers and prompt tuning
- Soft prompts and prefix tuning
- Merging fine-tuned adapters back into base models
- Running fine-tuning with the PEFT library in full

✅ **Strengths:** The authoritative technical reference for PEFT. Maintained by the HuggingFace team with runnable examples for every technique.

🎯 **Best for:** Deep technical implementation of efficient fine-tuning on your own hardware.

6.9 🚀 System Design

Building and Deploying Real Systems

📍 *Only makes sense after you have built real systems — APIs, LLM pipelines, RAG systems, agents. After Phase 6 you have built all of those. Now you learn to design them at scale. Before Phase 6 it is too abstract to stick.*

#1 — Best Overall — System Design Primer

donnemartin · GitHub · Beginner → Advanced · Self-paced

🔗 https://github.com/donnemartin/system-design-primer

What you will learn

- How to approach a system design problem from scratch
- Scalability fundamentals — vertical vs horizontal scaling
- Load balancers, reverse proxies, and API gateways
- Databases — SQL vs NoSQL, replication, sharding, indexing
- Caching strategies — CDN, Redis, cache invalidation patterns
- Asynchronous processing — message queues, Kafka, task workers
- CAP theorem — consistency, availability, and partition tolerance
- Microservices vs monolith — when each makes sense
- Designing real systems — URL shortener, Twitter feed, Google Drive, chat app

✅ **Strengths:** The single most comprehensive free system design resource available. Used by engineers preparing for FAANG interviews. Covers every concept from beginner to advanced with diagrams and real examples. No account needed.

🎯 **Best for:** Learning system design thoroughly from first principles

#2 — ByteByteGo YouTube Channel

YouTube · Beginner → Advanced · Ongoing

🔗 https://www.youtube.com/watch?v=lX4CrbXMsNQ&list=PLCRMIe5FDPsd0gVs500xeOewfySTsmEjf

What you will learn

- Visual breakdowns of every concept in the System Design Primer
- How real systems are built — Instagram, Netflix, Uber, WhatsApp
- Database internals — how indexes, transactions, and replication actually work
- API design patterns — REST, GraphQL, gRPC
- Distributed systems concepts — consistency models, leader election, consensus
- AI-specific system design — serving LLMs at scale, vector database architecture, RAG pipelines in production

✅ **Strengths:** The best visual companion to the System Design Primer. Every abstract concept becomes concrete with diagrams and animation. Alex Xu who runs the channel literally wrote the book on system design interviews.

🎯 **Best for:** Watching after each Primer section to make the concept fully click visually

#3 — Exponent YouTube Channel

YouTube · Intermediate → Advanced · Ongoing

🔗 https://www.youtube.com/watch?v=cL9If4X7aaE&list=PLrtCHHeadkHp92TyPt1Fj452_VGLipJnL

What you will learn

- Full mock system design interviews from start to finish
- How to structure your thinking when given a design problem
- How to communicate tradeoffs clearly under pressure
- Designing systems for scale — when to add caching, when to shard, when to use a queue
- Common interview mistakes and how to avoid them
- AI system design interviews — designing a recommendation engine, a search system, an ML platform ith Docker and cloud

✅ **Strengths:** The only free resource that shows you how system design actually looks in an interview setting. Watching real mock interviews teaches you the format and communication style that no textbook can.

🎯 **Best for:** fter finishing the Primer and ByteByteGo — seeing how to present and discuss designs in a real interview context.

GeeksForGeeks Course (all GFG phases): https://www.geeksforgeeks.org/batch/ds-16?tab=Resources

DeepLearning.AI Short Courses (Phase 6 permanent resource): deeplearning.ai/short-courses

GPU Access — Kaggle Notebooks (Free T4 GPU · 30hrs/week · use for Phase 5, 6.8, and ARENA): https://www.kaggle.com/code

GPU Access — Google Colab (Free T4 GPU backup · use when Kaggle quota runs out) *https://colab.research.google.com/*